

MATHEMATICS

Digital Economics and Business

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- Write first name, last name, and student number on any blank paper sheet.
- Exam duration: 1 hour.
- The use of calculators, phones, smartwatches, or any other type of material is not permitted.
- Write all results and procedures on the blank paper sheets.
- **Correct final results without a coherent and logically justified derivation receive zero credit**
- Once the exam has started, you cannot leave the room, unless:
 - you are willing to withdraw from the exam;
 - you finish the exam before the end.
- You can leave the class and hand in your test only 30 minutes after the start but not in the last 5 minutes of the exam.
- At the end of the test, hand in this exam sheet inside all draft papers.
- You are not allowed to bring any paper out of the room.

1. Study and graph the following function:

$$f(x) = e^{-x}\sqrt{x-1}$$

Check for the existence of possible non-differentiability points.

2. Compute and graph the area of the region included between the curves $y^2 = 9x$ and $x^2 = 9y$.

3. Find all the local maxima, local minima, saddle points of the following function:

$$f(x, y) = e^{x-y}(x^2 - 2y^2)$$

4. Let $A = \begin{pmatrix} 1 & -1 \\ 1 & k \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$. Find k such that $rk(AB) = 1$

5. Let $A = \begin{pmatrix} 2 & 1 & 3 & 0 \\ -1 & 0 & 1 & 2 \\ 2 & 0 & -1 & -1 \\ -3 & 1 & 0 & 1 \end{pmatrix}$. Compute $\det(A)$.
